

SCIENTIFIC REPORT

**SEVENTEENTH INDIAN EXPEDITION TO
ANTARCTICA**

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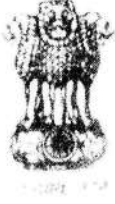
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FOREWORD

Since its inception in 1981, the Indian Antarctic Programme has evolved into a sustained scientific campaign with a clear-cut focus to understand the mysteries of nature which can be best appreciated from Antarctica. For the last forty years, the Antarctic Treaty has provided a stable framework for human activities in Antarctica, thus making available valuable scientific information for the benefit of posterity.

The successful conclusion of the 17th Indian Antarctic Expedition and the scientific results generated as reflected in this report, has further consolidated our knowledge in the various domains of Polar Science. This expedition has been remarkable in many ways and in particular in having a distinct thrust towards contributing to some of the SCAR mounted international campaigns by establishing the permanent seismic, GPS and VLF observatories at Maitri. Moreover, true to the spirits of the Antarctic Treaty, the 17th IAE saw the active participation of two scientists from Germany who collaborated with their Indian counterparts in establishing the GPs observatory at Maitri. Similarly, a strong complement of biodiversity experiments in this expedition shall help the biological community to appreciate the diversity and adaptability of the unique flora and fauna of the Schirmacher Oasis region of Antarctica.

The present report is an outcome of a creditable job performed by a multi-disciplinary and multi-institutional team under the capable leadership of Shri KR Sivan from R&DE Engineers, Pune. I am sure that this technical report published under the close supervision of Shri Sivan, will serve as a valuable source of data and information in the field of Polar Science and Logistics.


(A.E. Muthunayagam)



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DR. P.C. PANDEY

Director

18 September 2000.

PREFACE

The role of Antarctica as a global modulator influencing the world climate and ocean circulation pattern, which in turn sustains the very existence of human beings on this planet earth, is becoming apparent day by day. Obviously therefore, science becomes the principal human activity in Antarctica prompted by an inter-disciplinary and inter-institutional approach adopted with an international matrix.

The completion of the XVII Indian Antarctic Expedition is a reflection of fulfilling the above goals in appreciating and understanding the various enigmas of nature controlled by this remote continent. This expedition has a special significance in having been launched for the first time from the new premises of the National Centre for Antarctic & Ocean Research, Goa. It is, therefore, most heartening to note that this expedition has completed its assigned scientific and logistics tasks as reflected in this technical report.

I would like to put on record, my deep appreciation of the concerted efforts made by the entire team under the able guidance of the leader, Shri K.R. Sivan, in achieving the targets of this mission. New dimensions were added, both on the scientific and logistic fronts including elements of international cooperation, in this expedition. State of the art experiments on teleseismic studies, planetary geodetic studies, measurement of ozone using laser heterodyne system and VLF propagation studies initiated in this expedition, has provided valuable scientific information as reproduced in this report. Another major thrust in this expedition has been a number of new experiments on biodiversity studies of different ecological niches, contributing towards the SCAR BIOTAS programmes.

The publication of this technical report heralds the completion of yet another scientific mission of the country paving way for new challenges. I congratulate Shri K.R. Sivan in getting this report published with his untiring efforts and the cooperation received from R&DE Engineers, Pune, in the publication work.


(P.C. PANDEY)

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THE TEAM

Sr.No.	Name	Organisation
XV Winter Team		
1.	Sh KR Sivan Leader & Station Commander	R&DE (Engrs), DRDO
2.	Sh Arun Chaudhary	SASE, DRDO
3.	Sh IP Raju	NGRI
4.	Sh SS Bist	IIG
5.	Sh BN Vishnoi	IMD
6.	Sh Manmohan Singh	IMD
7.	Sh Shambhu Nath	NPL
8.	Sh A Prasad Goud	R&DE (E), DRDO
9.	Sh V. Nautiyal	DEAL, DRDO
10.	Sh S. Kaushik	DEAL, DRDO
11.	Dr. SK Sharma	DGHS
12.	Sh Kishan Lal	ITBP
13.	Maj G Narayan	Indian Army
14.	Sub KS Panesar	Indian Army
15.	Sub RS Aswal	Indian Army
16.	Nb/Sub BS Mor	Indian Army
17.	Hav K Madhavan	Indian Army
18.	Hmt Ratan Singh	Indian Army
19.	Hav Jahabar	Indian Army
20.	Hav Rajbir Singh	Indian Army
21.	Hav GD Reddy	Indian Army
22.	Hav NN Maharana	Indian Army
23.	L/Nk Devi Singh	Indian Army
24.	Nk Anil Kumar	Indian Army

Summer Team

1. Sh MJ D'souza	GSI
2. Sh A Kundu	GSI
3. Sh Amrik Singh	SASE, DRDO
4. Sh S. Bose	Jadhavpur University
5. Sh S. Samanta	Jadhavpur University
6. Sh V Ravichandran	SOI
7. Sh L Prem Kishore	NGRI
8. Dr. S H Jafri	NGRI
9. Sh D K Meshram	IIG
10. Dr. Lakha Singh	NPL
11. Sh NV Satyanarayan	NPL
12. Dr. A K Gwal	Baraktullah University
13. Dr. R P Barman	ZSI
14. Dr. J R Sharma	BSI
15. Sh S C Singh	BHU
16. Sh A N Sahi	BHU
17. Dr Vivek Pandey	NBRI
18. Sh S Gupta	NII
19. Sh S Sen	R&DE (E), DRDO
20. Sh B M Shaligram	R&DE (E), DRDO
21. Sh Kulbhushan Rai	R&DE (E), DRDO
22. Sh P Kundu	DEAL, DRDO
23. Sh I Rajasekhar	NAL
24. Sh Rakesh Lal	ITBP
25. HMT R V Ghatge*	Indian Army
26. Shri P S Singh	DOD
27. Dr. Wilfried Korth	IPG, Dresden, Germany
28. Mr. James Perl	IPG. Dresden, Germany
29. Mr. Lutz Eberlein	IPG. Dresden, Germany

* Member scheduled for winter returned after summer on medical grounds.



Expedition Members on the deck of MV Polar Bird



XV WOT Members



Guard of honour to Hon. Dy.P M. Mauritius Mr. Kailash Purryag

